

RACTF 2020 Crypto - 01

deuterium — 2020-06-09

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/ractf/crypto/01/2020-06-09-RACTF-2020-01.html>

Abstract

CTF writeup: RACTF 2020's 01 swaps the alphabet and misplaces the spaces. quipqiup solves the monoalphabetic substitution; word spacing still needs a human.

01

No information about the cipher is provided

```
LHFKM GMRHC FLMMJ ULXFY JOUFC
FQFXF ZJOKP JOMMU LMRJT FFTBA
JYFFR JZFXG AWJCB ULXFI FFKRF
KPGKH RFWCF MTFRR LHFRI FMQFF
KFLWU JMUFC IOMMU FYCFF KWCYB
MFPQF CFHJG KHMJK FFPMJ PFWCY
BMMUF TMJQJ CVJOM GZMUF CFRRJ
TFMUG KHGAA FHLAH JGKHJ KLKPH
FMMUF TLCCF RMFPK JCTLA YMUFW
CYBMJ HCLBU YMFLT QJOAP PJMUG
RIOMM UFYXF LAAPF WGPFP LMMUF
RLTFM GTFMJ HJJKL KOLAA FLXFI
FRMJZ AOWVL HFKMI MUFRR WCFMW
JPFGR PJWOT FKMR
```

We have to solve the cipher. Considering the level and monoalphabetic substitution i.e each alphabet is replaced with other alphabet.

Quickly verifying my hypothesis by checking on [quipqiup](#) I found the plaintext

```
AGENT ITS GREAT TO HAVE YOU HER E WEVE FOUND OUT THAT SOME EMPL OYEES OF EVIL CORP
HAVE BEEN SE NDING SECRET
MESSAGES BETWEE N EACH OTHER BUT THEYRE ENCRYP TED WERE GOING TO NEED TO DECRY PT THEM
TO WORK OUT IF THERES
SO METHING ILLEGAL GOING ON AND G ET THEM ARRESTED NORMALY THE C RYPTOGRAPHY TEAM
WOULD DO THI S BUT THEYVE
ALL DECIDED AT THE SAME TIME TO GO ON ANUAL LEAVE B EST OF LUCK AGENT B THE SECRET C
ODE IS DOCUMENTS
```

Correcting spaces a bit, we get the text [decrypted.txt](#)

```
AGENT ITS GREAT TO HAVE YOU HERE WEVE FOUND OUT THAT SOME EMPLOYEES OF EVIL CORP HAVE
BEEN SENDING
SECRET MESSAGES BETWEEN EACH OTHER BUT THEYRE ENCRYPTED WERE GOING TO NEED TO DECRYPT
THEM TO WORK
OUT IF THERES SOMETHING ILLEGAL GOING ON AND GET THEM ARRESTED NORMALY THE
CRYPTOGRAPHY TEAM WOULD
```

DO THIS BUT THEYVE ALL DECIDED AT THE SAME TIME TO GO ON ANUAL LEAVE BEST OF LUCK
AGENT B THE SECRET
CODE IS DOCUMENTS

Hence the code we are looking for is **DOCUMENTS**

Alternatively, you may look up the video writeup by @hyperreality on puzzles 01-06, they were quite informative, given that I got stuck at 04

[YouTube link](#)

| Interactive controls are omitted from this PDF. See the canonical HTML version.